Scalability & Future Roadmap

Template: Scalability Strategy and Long-Term Evolution Plan	**Maximum Marks: 1 Mark**
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Current System Limitations:

S. No	Limitation	Operational Impact	Priority to Address
1	Single-Camera RGB Input	Subsurface produce decay cannot be detected until surface discoloration occurs.	High
2	Cloud API Latency Dependency	Inference depends on internet bandwidth (1.5-2s latency).	Medium
3	Manual Video Upload for Audits	Kitchen CCTV currently ingested via batch uploads rather than live RTSP streams.	High

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single worker FastAPI instance handling ~100 concurrent requests.	Kubernetes auto-scaling pod clusters behind AWS ALB handling 50,000+ concurrent users.
2	Data Storage	Stateless ephemeral buffer with zero persistent client photo storage.	Encrypted PostgreSQL + Supabase vector store for historical nutritional analytics.
3	Performance	Cloud inference via Groq API taking ~1.8 seconds per plate scan.	On-device quantized TensorRT / ONNX micro-models executing in < 250ms.
4	Security & Auditing	Environment-variable token authentication.	OAuth 2.0 / JWT role-based access control with SOC-2 type II compliance.

Future Roadmap (Phase 2 to Phase 4):

Phase	Planned Feature / Enhancement	Target Timeline	Expected Strategic Impact
Phase 2	Native iOS & Android Mobile Apps with Real-Time Camera AR HUD Overlay	Q3 2026	Direct consumer adoption; instant real-time scanning in grocery stores.
Phase 3	Continuous CCTV RTSP Stream Ingestion for Commercial Kitchen Hygiene	Q4 2026	Automated enterprise compliance alerts sent to restaurant managers.
Phase 4	Integration with Smart Home Appliances (Samsung Family Hub, Smart Ovens)	Q1 2027	Autonomous inventory tracking and automatic oven temperature calibration.